FORENSICS IN DIGITAL SECURITY (FDS)

Digital Investigator

OBJECTIVES

- Define computer forensics
- Describe brief history of computer forensics
- Describe digital forensics today
- Describe cybercrimes
- Describe forensic investigations
- Describe roles of forensic investigators

FORENSICS SCIENCE

"The application of ***scientific methods and techniques*** to ***matters under investigation*** by ***a court of law.***"

- From Oxford

"Forensic science is the ***use of scientific methods or expertise*** to ***investigate crimes or examine evidence*** that might be presented in ***a court of law***."

- By *National Institute of Standards and Technology (NIST)*

FORENSICS SCIENCE

Identifiable raw material

Derived from physical sciences

Unified body of theory

Yields to research methods

Standardised tests

COMPUTER FORENSICS

"A methodical series of ***techniques and procedures for gathering evidence***, from ***computing equipment and various storage devices and digital media***, that can be ***presented in a court of law*** in a coherent and meaningful format."

—Dr. H.B. Wolfe

"Computer forensics as the discipline that combines elements of law and computer science to **collect and analyze data** from ***computer systems, networks, wireless communications, and storage devices*** in a way that is ***admissible as evidence in a court of law.***"

- US-CERT

COMPUTER FORENSICS

Raw material is an artefact

Techniques are not "discoveries"

No standardised tools

Lack of proper training

More of an "Art" than a "Science"

LOCARD'S EXCHANGE PRINCIPLE

"Every contact leaves a trace."

- Edmond Locard

Criminal always

Leaves some bit of evidence at the crime scene	Take some bit of evidence from the crime scene	No matter how small

BRIEF HISTORY OF COMPUTER FORENSICS

30 years before

Computer was not thought to be part of everyday life

1970

Electronic crimes increase, in financial sector – White collar fraud

1980s

Surged in PCs with different OS use in companies and government agencies

Mid-1980s

- Recovery tools appeared, Xtree Gold and Norton DiskEdit.
- Hardisk size grew from 10Mb to 60 Mb
- FBI Computer Analysis and Response Team (CART) emerged

1990s

- Specialised forensic tools
- Hardisk size grew from 1GB to 8GB
- Scientific Working Group on Digital Evidence (SWGDE) was established to develop standards

2000s

Many software company create forensic tools for investigation use

DIGITAL FORENSICS TODAY

Digital Forensics

Computer Forensics	Investigates data that can be retrieved from a computer's hard disk or other storage media
Network Forensics	Information about how a perpetrator or an attacker gained access to a network
Mobile Forensics	Investigates data that can be retrieved from a mobile handset's storage media
Multimedia Forensics	Investigates data that can be retrieved from a multimedia such as Closed-circuit television (CCTV), videos, audios, etc.
VM Forensics	Investigates data that can be retrieved about virtual machines
More	Data Analysis, Data Recovery, Database Forensics, Cloud forensics, etc.

COMPLEXITY OF DIGITAL FORENSICS

- Need to determine role of the computer in the particular incident, i.e. use as a tool or is the target
- Computer might be used as a tool and is also the target of the attack
- Computer might be used in the commission of a crime. Computer itself as a digital crime scene containing evidence

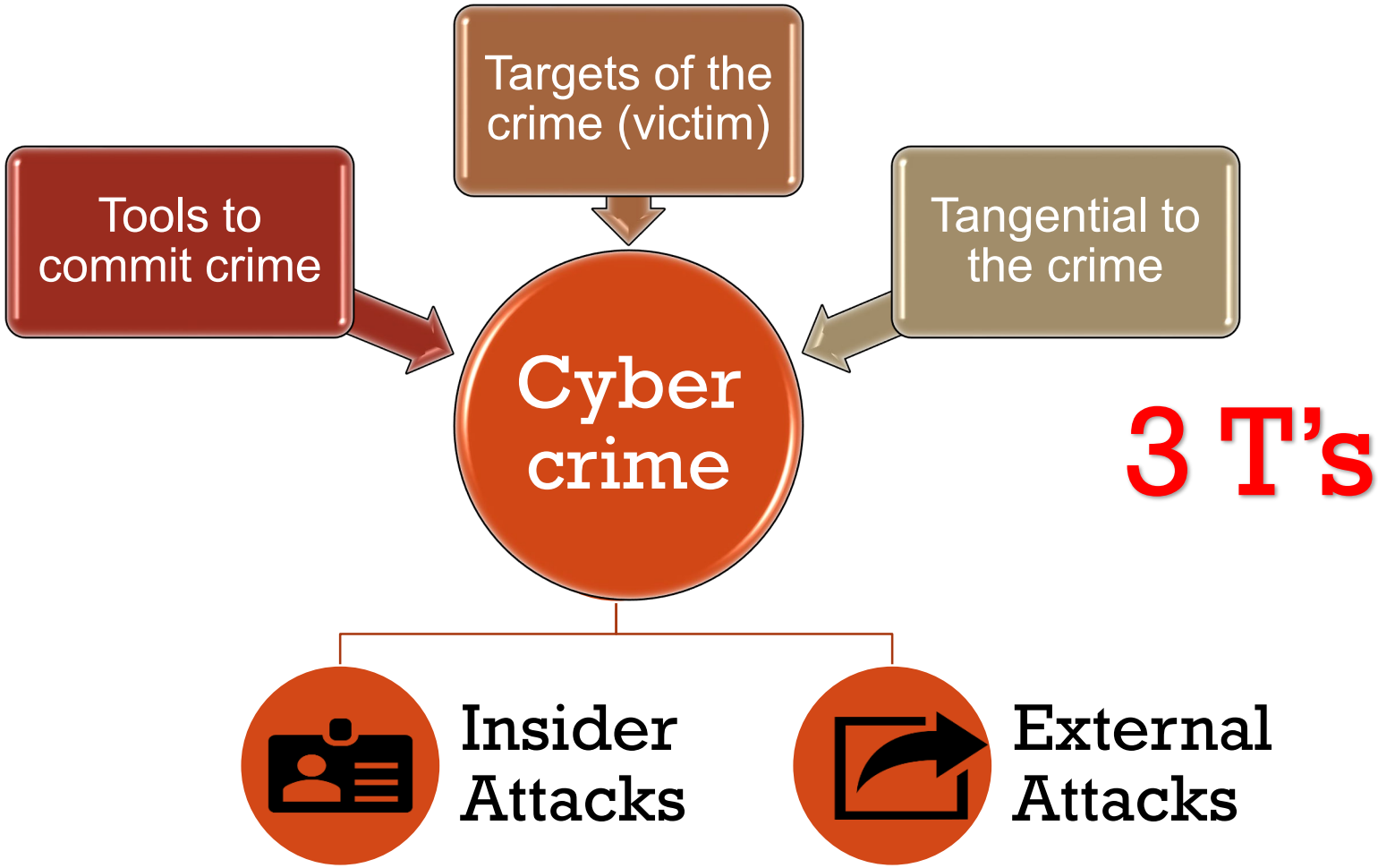
COMPLEXITY OF DIGITAL FORENSICS

- High speed and volumes
- Explosion of complexity
- Development of standards
- Privacy-preserving investigations
- Legitimacy
- Rise of anti-forensics techniques

Caviglione, Luca & Wendzel, Steffen & Mazurczyk, Wojciech. (2017). The Future of Digital Forensics: Challenges and the Road Ahead. IEEE Security and Privacy Magazine. 15. 10.1109/MSP.2017.4251117.

CYBERCRIMES

Crime must be intentional and not accidental



EXAMPLES OF CYBERCRIMES

- Theft of intellectual property
- Damage of company service networks
- Financial fraud
- Hacker system penetrations
- Denial of Service Attacks
- Distribution and execution of viruses and worms
- More

MOTIVATION OF CYBERCRIMES

- Experimentation of script kiddies to learn
- Psychological needs, e.g. o leave a mark
- Revenge and malicious reasons
- Desire to embarrass the target
- Espionage - corporate and governmental
- Disgruntled employee
- More

FORENSIC INVESTIGATIONS

Forensics investigators often work as part of a team, known as the **investigations triad**

Investigations Triad

(Adapted from Guide to Computer Forensics and Investigations)

RULES OF FORENSIC INVESTIGATIONS

- Examine original evidence least possible, examine duplicate evidence
- Follow rules of evidence, do not tamper evidence
- Prepare chain of custody, handle evidence with care
- Never exceed the knowledge base
- Document any changes to evidence

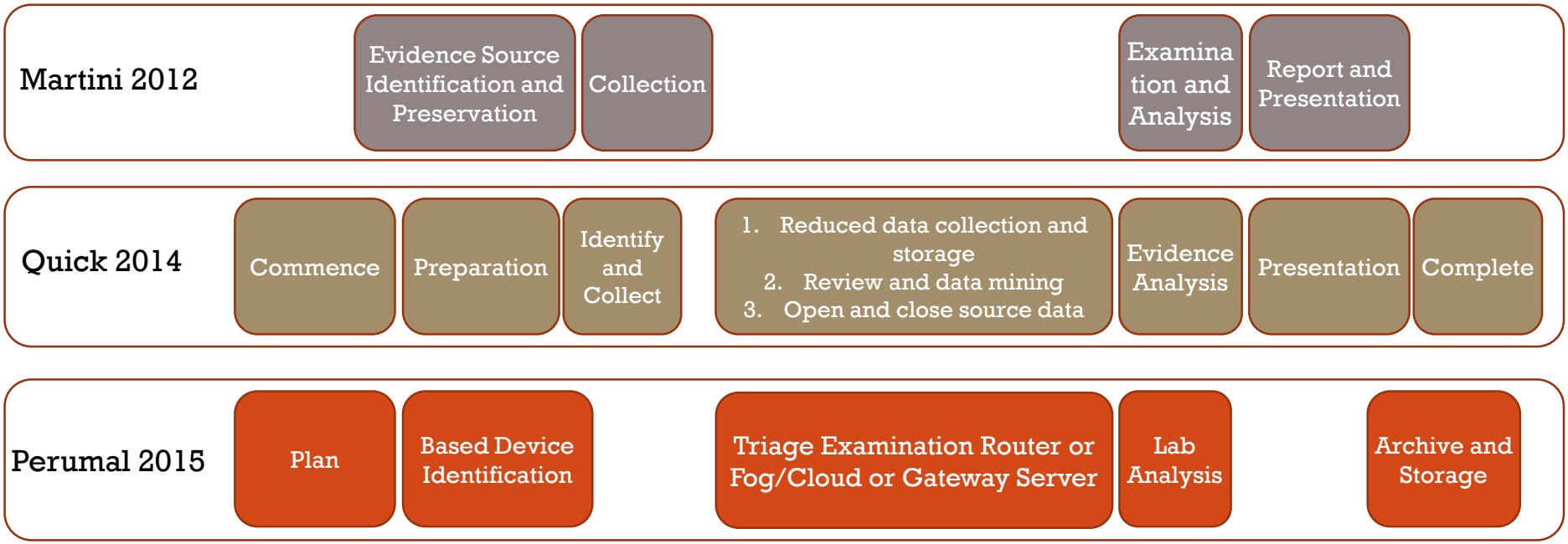
FORENSIC INVESTIGATIONS PROCESS

Identification	• Assessing the case to detect/identify the event/crime
Preservation	• Identify and preserve evidence with integrity • Maintain chain of custody when handling evidence
Collection	• Find evidence, discover data and gather evidence • Create hash of the evidence
Examination	• Trace, filter and extract hidden data
Analysis	• Analyse duplicate of evidence with forensic tools
Presentation	• Include results in investigation report • Expert witness in court
Decision	• Decision by final authorities like courts of law and corporate

RECENT DIGITAL FORENSIC MODELS

Some recent models

- An integrated conceptual digital forensic framework for cloud computing *(Martini & Choo 2012).*
- Data reduction and data mining framework *(Quick & Choo 2014).*
- Internet of Things (IoT) Based Digital Forensic Model *(Perumal et al. n.d. 2015)*



Adapted from Figure 3: Recent Digital Forensic Models for Handling Modern Advancements
Du, Xiaoyu & Le-Khac, Nhien-An & Scanlon, Mark. (2017). Evaluation of Digital Forensic Process Models with Respect to Digital Forensics as a Service.

CATEGORIES OF FORENSIC INVESTIGATIONS

Forensic Investigations

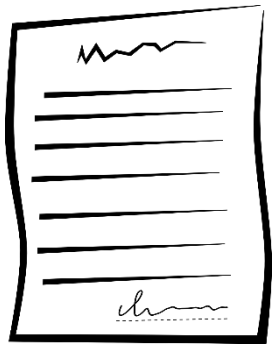
Public (Criminal)

- Involve government agencies
- Responsible for criminal investigations and prosecution
- Act on violation of state, federal, or international law

Private (Corporate)

- Involve entity of personal or corporate
- Act on possible violation of corporate regulations or personal interests

PUBLIC INVESTIGATION CASE FLOW



Complaint

- Any formal legal document that sets out the facts and legal reasons

Investigation

- Investigate Complaint
- Collect Evidence
- Build a Case

Prosecution

- Evidence Presented in Court
- Judge renders a judgement

Public-Sector Case Flow

(Adapted from Guide to Computer Forensics and Investigations)

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MORE

PUBLIC INVESTIGATIONS

Fourth Amendment (U.S. Constitution)	Affidavit
• *Protects everyone's rights* to be secure in their person, residence, and property. • *Search warrants* are *needed for search and seizure*	• *Sworn statement of support of facts* about or evidence of a crime • Submitted to a Registrar/District Judge/Magistrate *to request a search warrant*

MORE

CORPORATE INVESTIGATIONS

- E-mail harassment
- Falsification of data
- Sabotage
- Embezzlement
- Gender and age discrimination
- Industrial espionage
- More

Corporate investigations can turn into criminal investigations

MORE

CORPORATE INVESTIGATIONS

Establish Line of Authority

- Establish Company Policies
- Displaying Warning Banners

Prepare for Investigation

- Designate an authorized requester to conduct investigations
- Conduct security investigations to distinguish company's problem or criminal problem
- Distinguish personal and company property

Silver-platter doctrine

- Handing results of private investigations over to law enforcement because of indications of criminal activity

MORE

DIGITAL FORENSICS SOURCES

Computer	Storage media	CCTV
Social media	Email	Cloud
	More	

Question: Could the forensics examiner access the web data across the national border without the user consent ?

PROFESSIONAL CONDUCT OF FORENSIC INVESTIGATORS

Credibility

- Continual training
- Membership in professional organizations
- Maintaining objectivity

Ethical and Integrity

- Achieve a high public and private standing
- Maintain highest level of honesty, ethics and integrity

Confidentiality

- Only discuss cases at hand only with person who has the right to know

CONCLUSION

- Forensic Science and Computer Forensics
- Digital Forensics (Past and Present)
- Cybercrimes with Motivation
- Forensic Investigations
  - Rules of Forensics
  - Investigation Process
  - Public and Private Investigation
- Digital Forensics Sources
- Professional Conduct of Forensic Investigators

REFERENCES

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- Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.
- Carrier, B. (2005). *File System Forensic Analysis.* Addison-Wesley Professional.
- Messier, R. (2015). Operating System Forensics. Syngress.

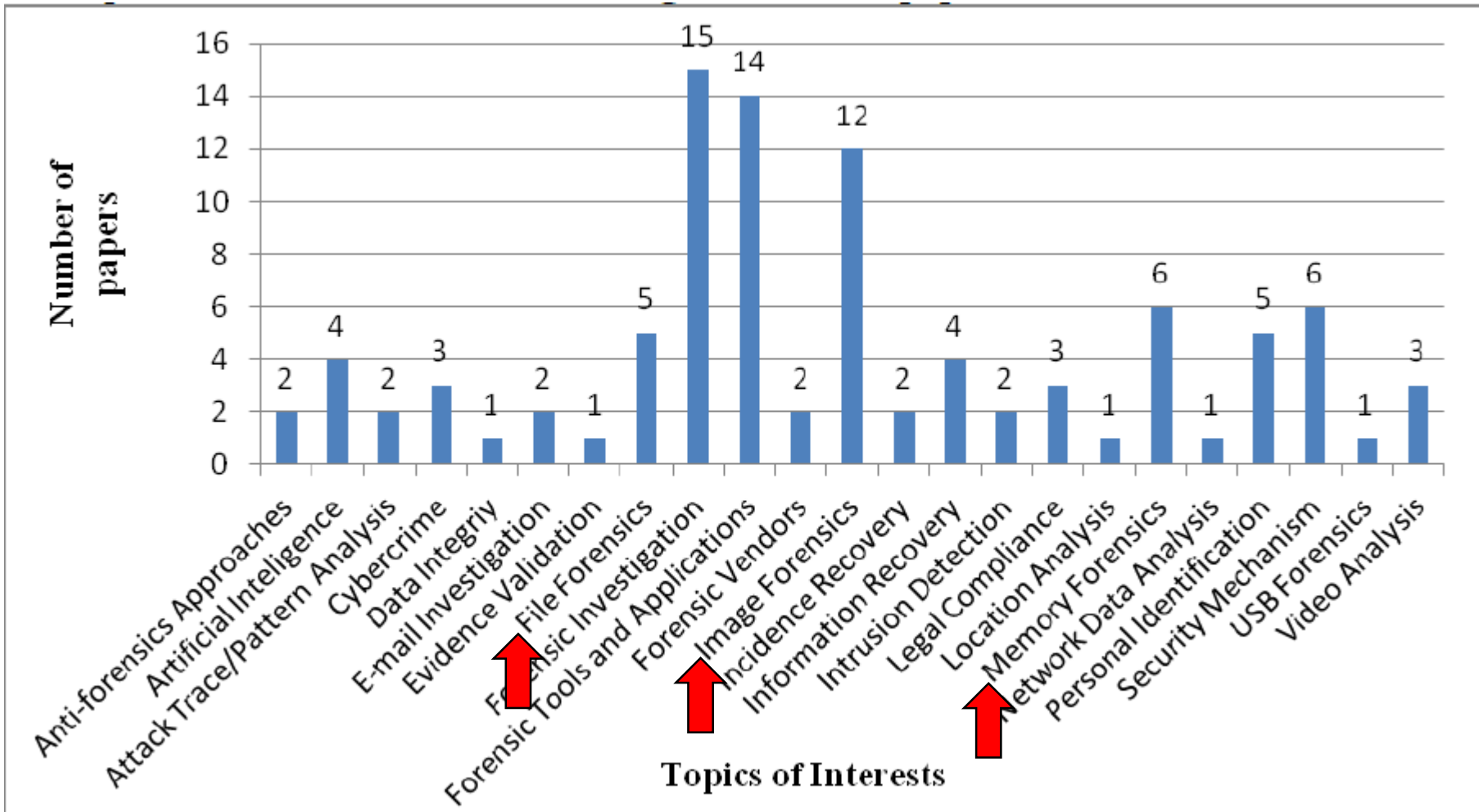
EXTRA SLIDES

UNDERSTANDING LAW ENFORCEMENTS AGENCY INVESTIGATIONS

- In a criminal case, a suspect is tried for a criminal offense, e.g. murder
- Computers and networks are only tools that can be used to commit crimes in traditional crimes.
  - E.g. "Cheating involving E-Commerce",
  - Scams such as "Credit-for-Sex" and "Internet Love".
  - Read "Singapore Police Force - MID-YEAR CRIME BRIEF FOR JANUARY TO JUNE 2015" .
- Singapore law on Computer Misuse and Cybersecurity Act, Chapter 50A. Read "Computer Misuse and Cybersecurity Act"
  - What happens if the victims were in Singapore while the bad guys were in other countries?

BACK

FROM ACADEMIC POINT OF VIEW IN DF



97 articles from 31 journals (from 2008 till 2013)

Source: Digital Forensic Trends and Future.
International Journal of Cyber-Security and Digital Forensics (IJCSDF) 2(2): 48-76

UNDERSTANDING CASE LAW

- Technology is evolving at an exponential pace
  - Existing laws and statutes can't keep up change
- Case law used when statutes or regulations don't exist
- Case law allows legal counsel to use previous cases like the current one
  - Because the laws don't yet exist
- Each case is evaluated on its own merit and issues

BACK

UNDERSTANDING LAW ENFORCEMENTS AGENCY INVESTIGATIONS (CONTINUED)

- Following the legal process Criminal case follows three broad stages
  - The complaint, the investigation, and the prosecution

Complaint → Investigation → Prosecution

Public-Sector Case Flow

(From Guide to Computer Forensics and Investigations)

BACK

UNDERSTANDING LAW ENFORCEMENTS AGENCY INVESTIGATIONS (CONTINUED)

- Following the legal process (continued)
  - A criminal case begins when someone finds evidence of an illegal act
  - Complainant makes an **allegation**, an accusation or supposition of fact
  - A police officer interviews the complainant and writes a report about the crime
    - **Police blotter** provides a record of clues to crimes that have been committed previously
  - Investigators delegate, collect, and process the information related to the complaint

BACK

UNDERSTANDING LAW ENFORCEMENTS AGENCY INVESTIGATIONS (CONTINUED)

- Following the legal process (continued)
  - After you build a case, the information is turned over to the prosecutor
  - **Affidavit**
    - Sworn statement of support of facts about or evidence of a crime
      - Submitted to a Registrar/District Judge/Magistrate to request a search warrant
    - Have the affidavit notarized under sworn oath – by the Commissioner of oaths & Public notary
  - Registrar/District Judge/Magistrate must approve and sign a search warrant
    - Before you can use it to collect evidence

BACK

UNDERSTANDING LAW ENFORCEMENTS AGENCY INVESTIGATIONS (CONTINUED)

Date ____ ____ ____

Based on actual inspection of spreadsheets, financial records, and invoices, Joe Smith, a computer forensics expert, is aware that computer equipment was used to generate, store, and print documents used in Jonathon Douglas's tax evasion scheme. There is reason to believe that the computer system currently located on Jonathon Douglas's premises is the same system used to produce and store the spreadsheets, financial records, and invoices, and that both the [spreadsheets, financial records, invoices] and other records relating to Jonathon Douglas's criminal enterprise will be stored on Jonathon Douglas's computer.

Source: *Searching and Seizing Computers and Obtaining Electronic Evidence in Electronic Investigations*, U.S. Department of Justice, July 2002.

Typical Affidavit Language

From Guide to Computer Forensics and Investigations

BACK

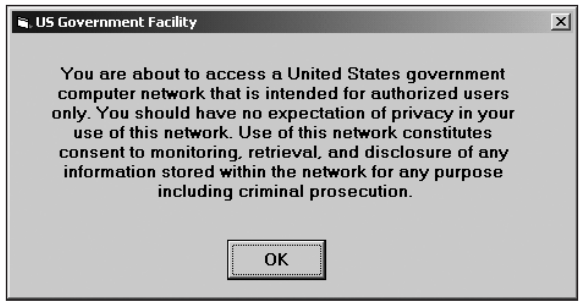
CORPORATE INVESTIGATIONS

- Establishing company policies
  - One way to avoid litigation is to publish and maintain policies that employees find easy to read and follow
  - Published company policies provide a line of authority
    - For a business to conduct internal investigations
  - Well-defined policies
    - Give computer investigators and forensic examiners the authority to conduct an investigation

BACK

CORPORATE INVESTIGATIONS

- Displaying Warning Banners
  - Usually appears when a computer starts or connects to the company intranet, network, or virtual private network
  - Informs end users that the organization reserves the right to inspect computer systems and network traffic at will
  - Make sure company displays well-defined warning banner
  - Another way to avoid litigation
  - Establishes the right to conduct an investigation



Sample Warning Banner

(From Guide to Computer Forensics and Investigations)

BACK

CORPORATE INVESTIGATIONS

- Designating an authorized requester
  - Authorized requester has the power to conduct investigations
  - Policy should be defined by executive management
  - Groups that should have direct authority to request computer investigations
    - Corporate Security Investigations
    - Corporate Ethics Office
    - Corporate Equal Employment Opportunity Office
    - Internal Auditing
    - The general counsel or Legal Department

BACK

CORPORATE INVESTIGATIONS

- Conducting security investigations
  - Types of situations
    - Abuse or misuse of corporate assets
    - E-mail abuse
    - Internet abuse
  - Be sure to distinguish between a company's abuse problems and potential criminal problems
  - Corporations often follow the silver-platter doctrine
    - What happens when a civilian or corporate investigative agent delivers evidence to a law enforcement officer

BACK

CORPORATE INVESTIGATIONS

- Distinguishing personal and company property
  - Many company policies distinguish between personal and company computer property
  - One area that's difficult to distinguish involves PDAs, cell phones, and personal notebook computers
  - The safe policy is to not allow any personally owned devices to be connected to company-owned resources
    - Limiting the possibility of commingling personal and company data

BACK

DIGITAL FORENSICS SOURCES

- Computer – Desktop computer, laptop, Server, Datacenter Server
- Other storage media – digital camera, thumb drive, SD Card, DVD or CD
- There are lots of digital evidence in the crime scene.
  - Evidence can be inculpatory ("incriminating") or exculpatory
  - Do we recover every single digital evidence?
  - Scope of search
- Read Page 1 to 12 from "A Guide for First Responders"

DIGITAL FORENSICS SOURCES

- CCTV
  - Closed-Circuit Television, Video format, Resolution, Encode/Decode:
  - Singapore standard of CCTV
  - Read "VIDEO SURVEILLANCE SYSTEM (VSS) STANDARD FOR BUILDINGS"
- Social media
  - Facebook, WeChat, Twitter
  - Challenge: How to forensically collect artifacts from local or Server?
- Email
  - Gmail, Hotmail, Corporate Exchange
  - Challenge: How to forensically collect email (pst, eml) from local or Server?
- Cloud
  - iCloud, Google Drive
  - Challenge: How to forensically collect artifacts from local or Server?
- Question: Could the forensics examiner access the web data across the national border without the user consent ?